→ Introduction to numpy!

Numpy is a library of python.

The Origin of Numpy!

Note:- Python was always very powerful with maths.
Heigh Math computational power in python but one thing was missing in python and that was array.

Python has some inbuilt data structure like Lists, tupples, Dictionary, sets, but there was no array, so due to this reason in 1995 Jim Hugunin, creates the Numeric library.
This library provide array support. But the problem was performance was not good. So resolving this issue in the early's 2000 another library was developed, named Numarray.
So in 2005
Travis Oliphant A professor and Scientist Merged Numeric and Numarray and Named it Numpy.
So finaly arrays are in numpy

→ Numpy!

Now do you know how numpy was made.
Numpy is written in C which are compiled and superfast.
This is the main reason Numpy is up to 50x faster than python list.

→ So in numpy we will learn

1) Numpy array 2) Numpy Indexing
3) Numpy Operations 4) Numpy Exercise

→ Numpy Arrays:

→ First let's Understand what is Arrays
Note: in Python there is a list datastructure
[1, 2, 3, 4] → This is a python list.

and arrays is [1, 2, 3, 4]

So what's the difference:

→ Numpy Arrays!
when we were doing python we had the data structure list & Numpy is famous for arrays.

⇒ Difference:

Features	Pythonlist	Numpy Array
Datatype	Mixed allowed (Hetro-Genius)	Same data type Only (Homogens)
Performance	Slower	Much faster (c-level)
More efficiency	low	High.
Vector operations	Manual loops	Easy + fast

→ first we will learn in array:

1) Basic Array Creation

2) Array generation function

3) Random Generation function.

4) Array Attributes.

5) Array methods

6) Reshaping and resizing.